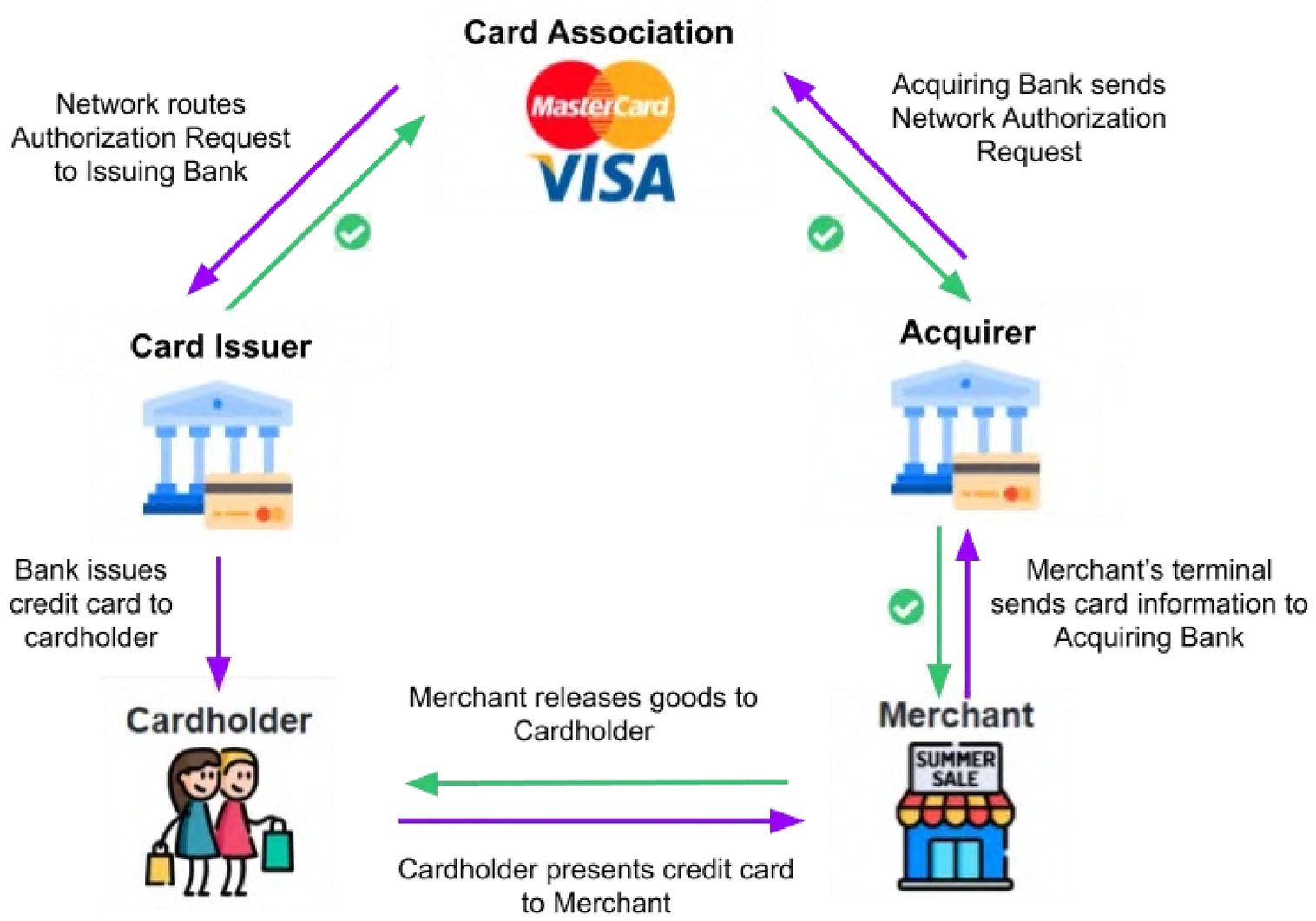
Blockchain



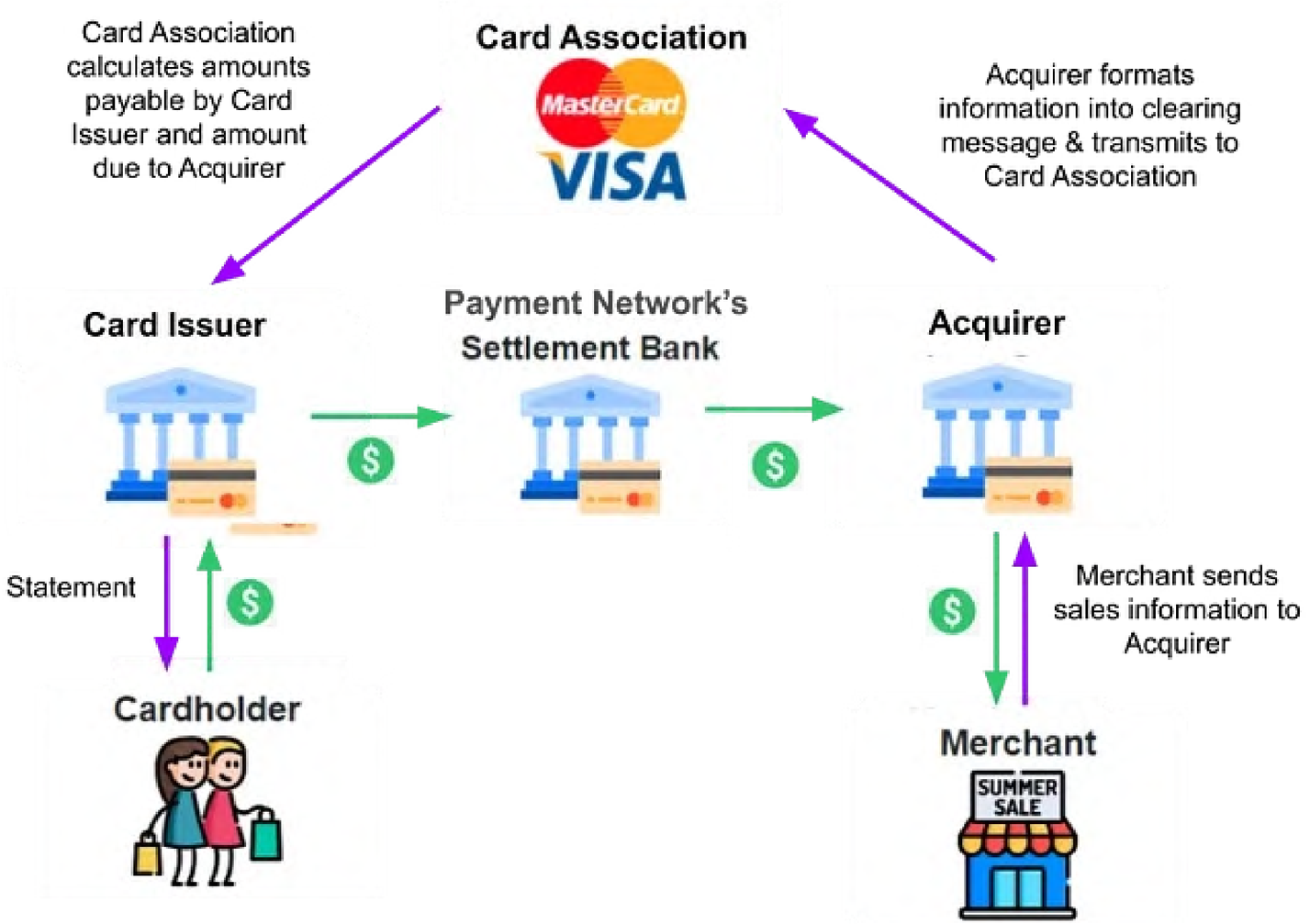
DRIM

Digital Payments

Credit Card Authorization

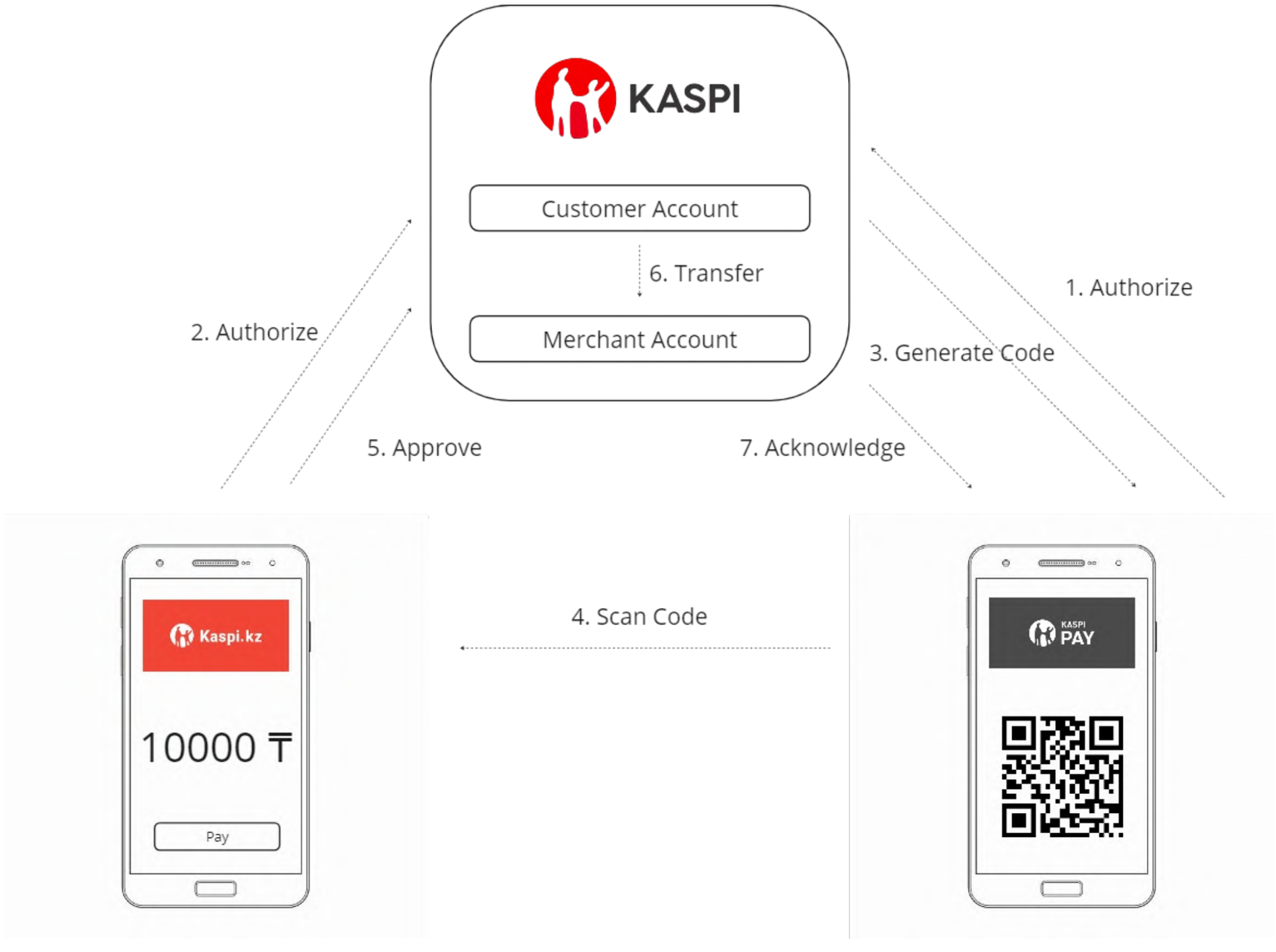


Credit Card Settlement



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Kaspi Pay



Ledger

A ledger is a record-keeping tool used to track financial transactions. It serves as a central repository for organizing and summarizing financial information.

Ledgers can be maintained manually or electronically, depending on the complexity of the transactions and the size of the organization.

Transaction 1: Deposit -> $100 -> John

Transaction 2: John -> $52 -> Michael

Transaction 3: Michael -> $10 -> Sam

Transaction 4: Sam -> $10 -> Withdraw

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Distributed Ledger

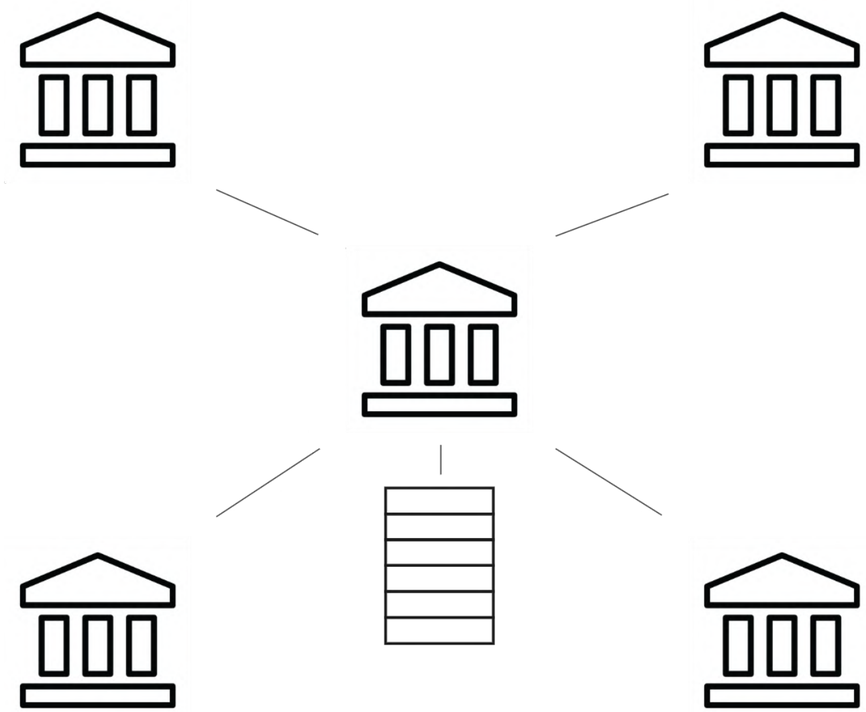
Distributed Ledger

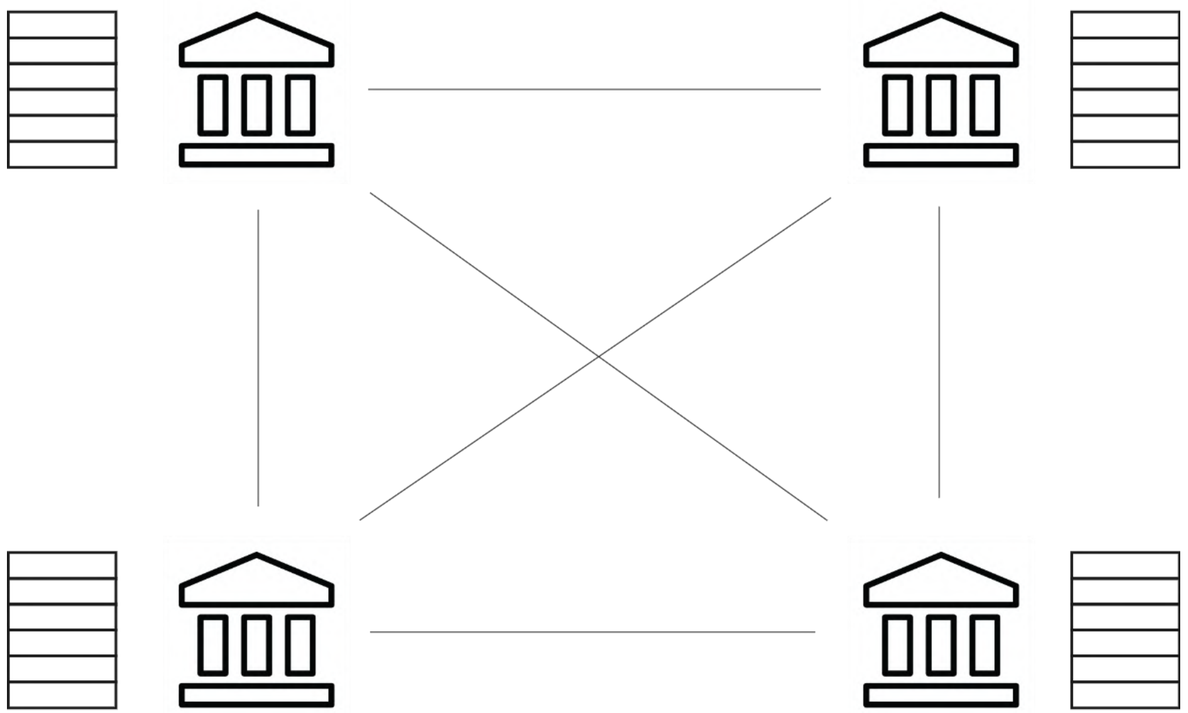
A distributed ledger is a type of database that is spread across multiple locations or nodes. It records and maintains a decentralized and shared history of transactions, contracts, or any other form of digital asset.

The ledger is collectively maintained and updated by a network of participants, often referred to as nodes or validators, rather than relying on a central authority.

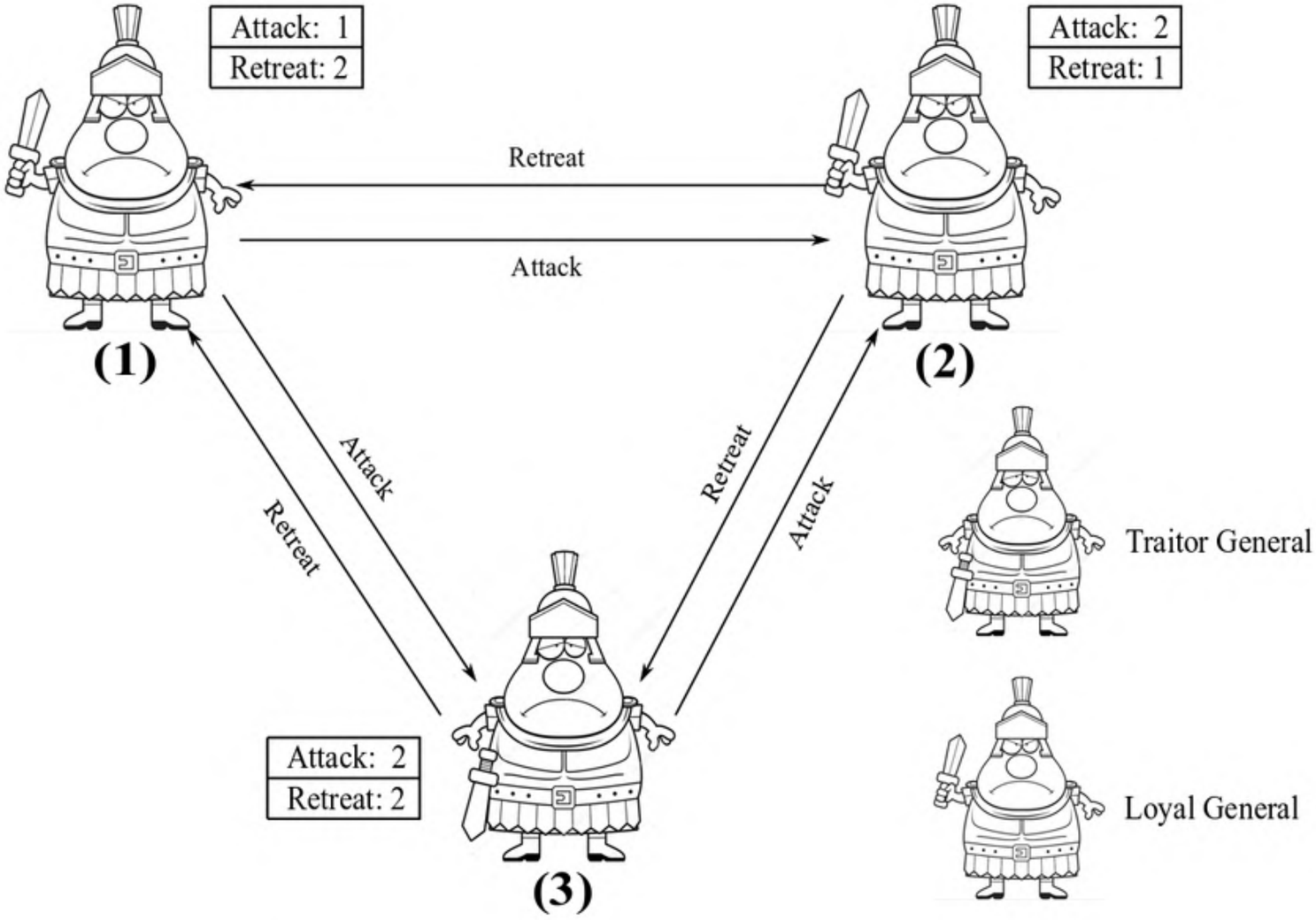
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Centralized vs Distributed Ledger





Byzantine Generals Problem



Distributed Consensus

Distributed consensus refers to the process of reaching an agreement among a group of participants in a distributed system, where each participant has its own copy of data or state.

This agreement is typically reached through a consensus protocol that ensures all participants agree on a certain value or state despite the presence of faults or malicious behavior.

Byzantine Generals Problem

=

Distributed Consensus Problem

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Blockchain

Satoshi Nakomoto

1. In August 2008, registered domain bitcoin.org
2. In October 2008, published the Bitcoin white paper titled "Bitcoin: A Peer-to-Peer Electronic Cash System"
3. On January 9th, 2009, released version 0.1 of the Bitcoin software on SourceForge, and launched the network by defining *the genesis block* of Bitcoin (block number 0)
4. Continued to collaborate with other developers on the Bitcoin software until mid-2010, making all modifications to the source code himself
5. Gave control of the source code repository to Gavin Andresen and vanished

Bitcoin Whitepaper

"Bitcoin: A Peer–to–Peer Electronic Cash System"

by Satoshi Nakamoto

https://bitcoin.org/bitcoin.pdf

Bitcoin Core

Bitcoin Core is a reference implementation of the Bitcoin protocol, which serves as the underlying technology behind the popular cryptocurrency, Bitcoin.

Developed by the Bitcoin Core project, Bitcoin Core is an open-source software project that provides the necessary software for users to interact with the Bitcoin network.

https://github.com/bitcoin/bitcoin

Blockchain vs Bitcoin

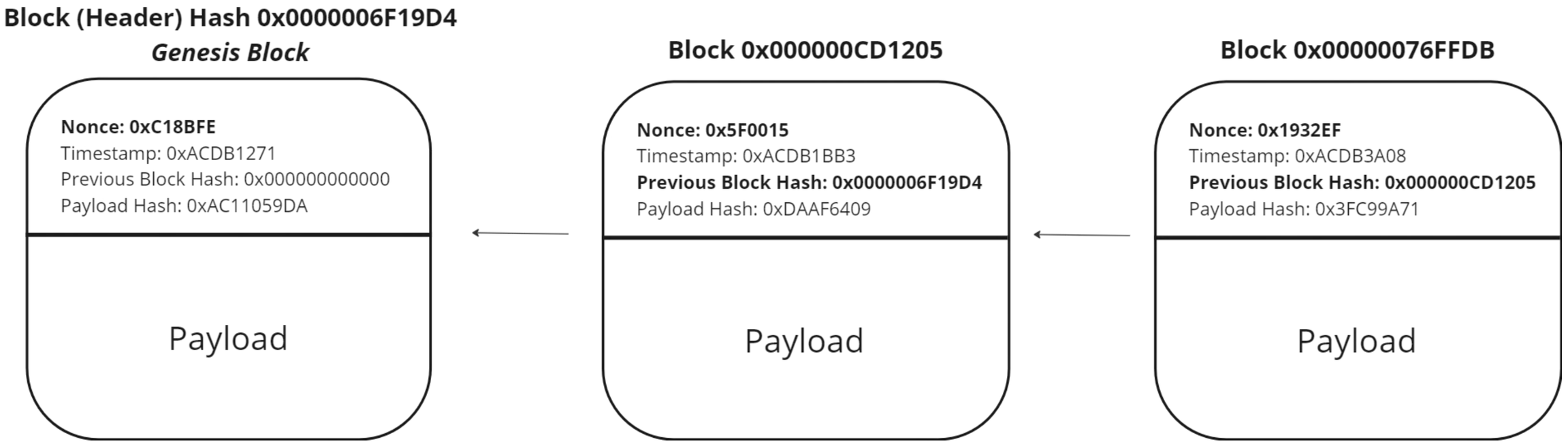
Blockchain and Bitcoin are related concepts, but they are not the same thing.

- A blockchain is a decentralized and distributed ledger technology that records transactions across multiple computers or nodes. It enables secure and transparent transactions without the need for a central authority.

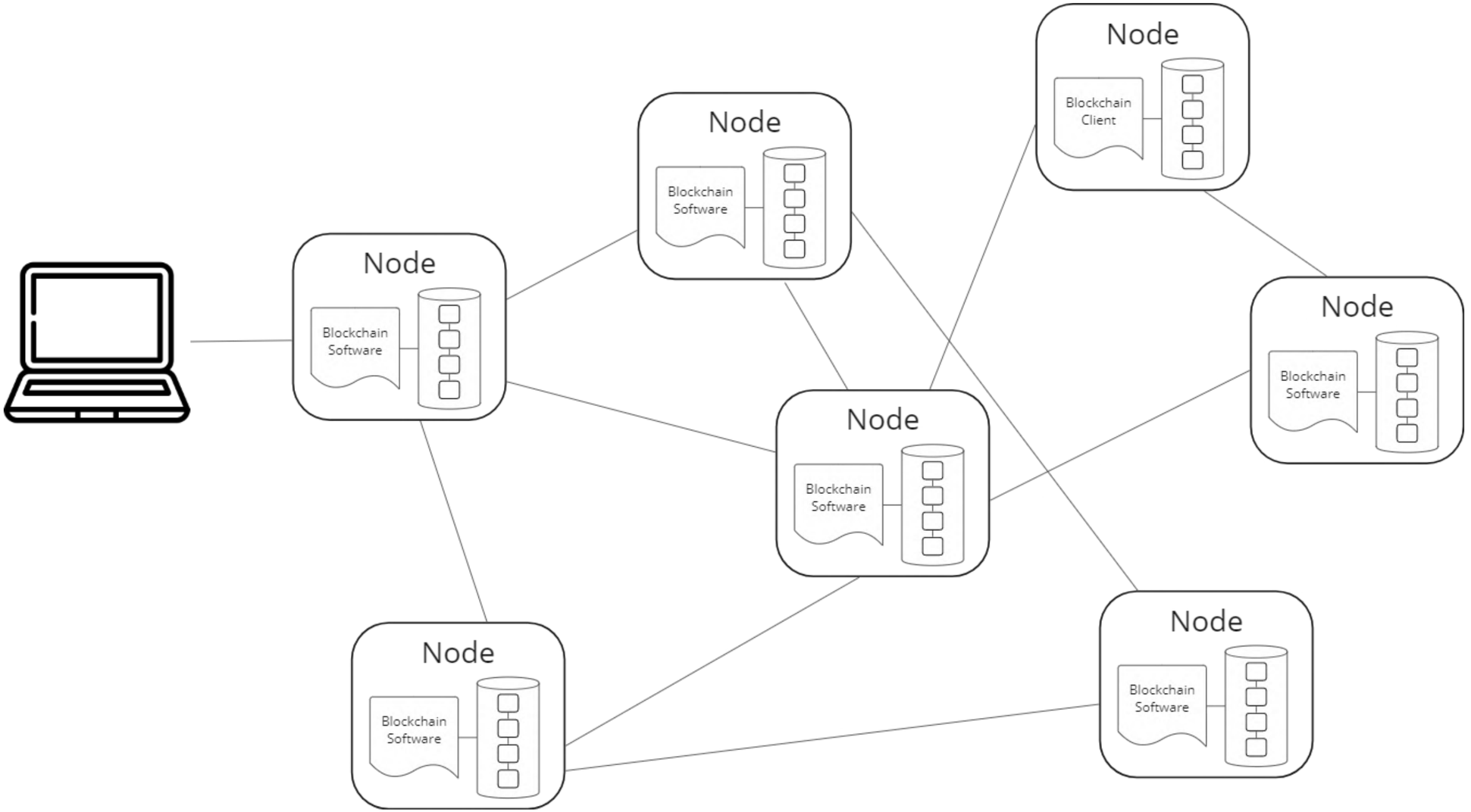
- Bitcoin, on the other hand, is a cryptocurrency and the first successful implementation of blockchain technology.

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Blockchain



Peer-to-Peer Network

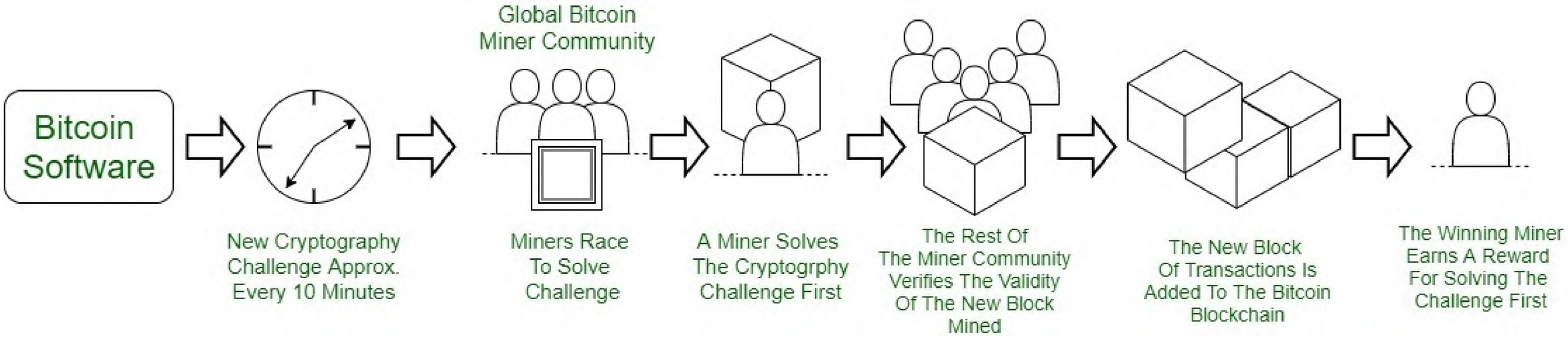


Mining

1. Receive and store broadcasted transactions
2. Select transactions and create a block with them
3. Calculate a payload hash, set timestamp
4. **Set a previous block hash**
5. Calculate a block header hash
6. If the block hash starts with N zeros, broadcast the block and go to 1
7. **Increment the nonce**
8. If new block received, go to 1
9. Go to 5

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Mining


Bitcoin Software
New Cryptography Challenge Approx. Every 10 Minutes
Global Bitcoin Miner Community
Miners Race To Solve Challenge
A Miner Solves The Cryptogrphy Challenge First
The Rest Of The Miner Community Verifies The Validity Of The New Block Mined
The New Block Of Transactions Is Added To The Bitcoin Blockchain
The Winning Miner Earns A Reward For Solving The Challenge First

Mining

Block hash calculation is extremely hard. It consumes a lot of CPU and electricity

But

Checking the block hash is instantaneous – just hash the block header and compare

This inequality is the foundation of blockchain safety and is the solution to the distributed consensus problem

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Trust

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Trust

Traditional money relies on trust

Blockchain is trustless

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Thank you! Questions?

